Muzly: A Multimodal Context-Aware Music Recommendation System Using CNN and CLIP

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***Abstract*—The rapid expansion of digital music streaming has created a growing demand for more personalized and adaptive recommendation systems. Traditional approaches, such as collaborative filtering, primarily rely on static listening histories and often fail to capture a user's immediate emotional and environmental context. This work presents the design and implementation of *Muzly*, a multimodal context-aware music recommendation platform. The proposed system integrates a Convolutional Neural Network (CNN) for real-time facial emotion recognition, OpenAI's CLIP model for zero-shot visual scene understanding, and external APIs to capture environmental context such as weather and geolocation. These heterogeneous inputs are fused through a weighted recommendation engine to generate dynamic, mood-aligned playlists. Experimental validation using a Streamlit-based web interface indicates that the platform supports real-time inference with stable performance and enhanced recommendation relevance, effectively addressing the limitations of conventional uni-modal music recommendation systems.**

***Index Terms*—Music Recommendation, Facial Emotion Recognition, CNN, CLIP, Context-Aware Computing, Multimodal AI, Streamlit.**

I. Introduction

Music is an intrinsic part of human culture, serving as a powerful medium for expression and emotional regulation. With the advent of digital streaming platforms, users have access to millions of songs; however, the challenge has shifted from access to discovery. Traditional recommendation systems predominantly rely on Collaborative Filtering (CF) or Content-Based Filtering (CBF). While effective at analyzing listening history, these methods suffer from the "cold start" problem and, more importantly, fail to account for the user's immediate psychological state or physical environment [1].

Most existing affective computing solutions are limited to a single modality—typically facial expression analysis. As observed in recent literature, these systems often perform poorly under varying lighting conditions and ignore broader contextual cues [2]. A user sitting in a dark room feeling "neutral" requires a different musical selection than a user in a sunny park feeling the same emotion. Currently, there is no unified framework that seamlessly integrates facial emotion, visual scene context (e.g., gym, party, nature), and environmental factors (weather, location) to provide a holistic recommendation.

This paper presents Muzly, an end-to-end multimodal recommendation system designed to support secure biometric analysis, real-time context understanding, and dynamic playlist generation. The platform integrates a CNN-based emotion detector, a CLIP-based scene classifier, and a Streamlit-based dashboard for operational monitoring.

II. Literature Survey

The music recommendation ecosystem spans affective computing, deep learning, and context-aware systems.

A. Deep Learning and Facial Emotion Recognition (FER)

A significant portion of existing research focuses on using Convolutional Neural Networks (CNNs) to detect emotions. Joseph et al. [1] developed an emotion-based system using Keras and CNNs to classify emotions and redirect users to YouTube playlists. Similarly, Guthula and Bordoloi [4] utilized the FER2013 dataset to train a model capable of recognizing seven emotions with reported competitive accuracy on the FER2013 dataset, using Haar Cascades for face detection. Variations of CNN architectures have also been explored; Deore [9] proposed a system integrating a Fuzzy Classification layer to refine detected emotions, while Angkasa and Maulidevi [10] combined convolution layers with XGBoost to address "cold start" issues in new user profiles.

B. Hybrid and Collaborative Filtering

Many systems rely on hybrid algorithms to improve accuracy. Wang [3] proposed a solution combining Collaborative Filtering (CF) with Tag-based methods using a "dynamic Jaccard coefficient" to capture changing preferences. Malleswari et al. [6] utilized a hybrid approach combining CF with "music gene" algorithms, minimizing error rates (RMSE) compared to traditional methods. Zeng [5] further refined personalization by combining Look-alike algorithms with K-means clustering.

C. Multimodal and Context-Aware Systems

Recent studies have begun to explore multimodal inputs. Chandraumakantham et al. [2] proposed a system leveraging Large Language Models (LLMs) trained on audio-visual features. Contextual factors like shared environments were investigated by Vitório and Silla [7], who used Graph Databases to model group preferences in settings like gyms or parties.

Sasaki et al. [8] pioneered an approach recommending music based on the mood of an input image rather than just facial expressions.

D. Research Gap

Despite these advancements, existing solutions exhibit limitations:

- **Uni-modal Dependency:** Heavy reliance on facial input leads to performance drops under poor lighting.
- **Lack of Environmental Context:** Factors like weather, location, and time of day are frequently ignored.
- **Rigidity:** Fixed input structures fail to handle diverse real-time scenarios.

Muzly addresses these gaps by fusing facial emotion (CNN), scene understanding (CLIP), and environmental data into a unified engine.

III. Methodology

The proposed methodology employs a cloud-native, service-oriented model comprising data preprocessing, parallel feature extraction, and a weighted recommendation engine.

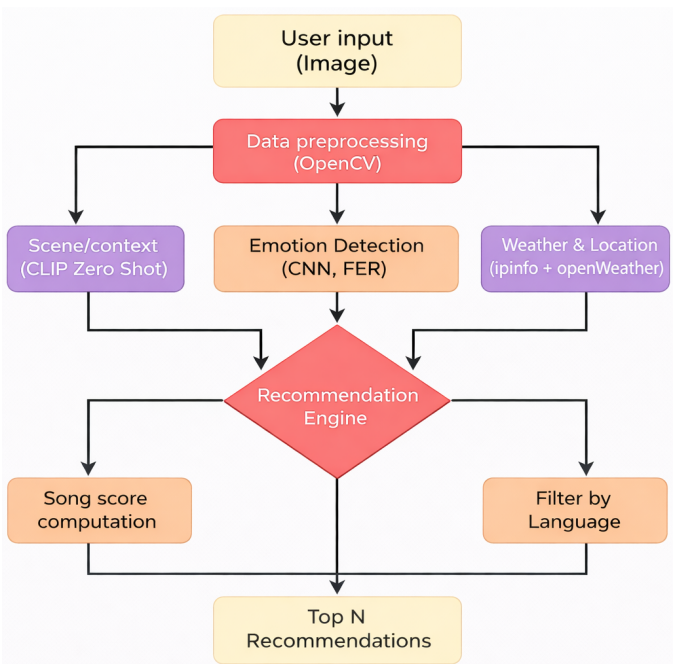

Fig. 1. High-Level System Architecture of Muzly.

A. Data Preprocessing

- **Face Detection:** The system utilizes the Haar Cascade classifier (`haarcascade_frontalface_default.xml`) to identify and crop faces from the input image.
- **Normalization:** Cropped face images are converted to grayscale and resized to 48×48 pixels. Pixel intensities are normalized to $[0, 1]$ to accelerate model convergence.
- **Augmentation:** The training dataset (FER2013) is augmented using ImageDataGenerator with rotation ($\pm 15°$), zoom (10%), and shear to prevent overfitting.

B. Parallel Feature Extraction

The system processes user input through three parallel streams:

1) **Facial Analysis:** Detects faces and classifies emotion using the trained CNN.
2) **Visual Context:** Analyzes the full scene using the CLIP model (`clip-vit-base-patch32`) to identify settings (e.g., "party," "office") and lighting conditions.
3) **Environmental Data:** Fetches real-time location via IPinfo and weather via OpenWeatherMap API.

C. Weighted Recommendation Engine

The final recommendation is generated using a hybrid filtering algorithm that calculates a Match Score (S) for every song. The scoring equation is defined as:

$$S_{total} = w_e \cdot M_{emotion} + w_l \cdot M_{language} + w_c \cdot M_{context} + w_w \cdot M_{weather} \tag{1}$$

Where:

- $w_e = 40$ (Emotion Weight)
- $w_l = 30$ (Language Weight)
- $w_c = 20$ (Context Weight)
- $w_w = 10$ (Weather Weight)

IV. System Architecture

The Muzly architecture is designed to support scalability and real-time interaction (see Fig. 1).

A. High-Level Design

The system integrates a Client Layer (Streamlit UI), an Application Layer (Python/Logic), and a Data Layer (CSV/APIs).

- **Input:** Users provide an image or video stream via the frontend.
- **Processing:** The backend uses TensorFlow for emotion inference and HuggingFace Transformers for scene analysis.
- **Output:** The dashboard displays the detected mood, scene, weather, and the recommended song list.

B. Frontend Dashboard

The frontend is implemented using Streamlit, providing a responsive interface for users. It supports image upload, camera integration, and displays dynamic "Song Cards" with play/like buttons. Custom CSS is used to implement a dark-themed, immersive UI.

C. Backend Services

The backend operates on Python:

- **Emotion Module:** Loads the `.h5` Keras model to predict one of 7 emotions.
- **Context Module:** Uses CLIP to compute cosine similarity between the input image and text descriptions (e.g., "sunny day," "night time").
- **Integration Module:** Handles API calls to OpenWeatherMap and IPinfo to fetch metadata.

V. Implementation

The implementation transforms the architectural design into a functional system capable of real-time analysis.

A. Facial Emotion Recognition (FER) Model

The CNN was trained on the FER2013 dataset (35,887 images). The architecture consists of three convolutional blocks, each containing two Conv2D layers, Batch Normalization, Max Pooling, and Dropout ($0.25 - 0.3$). The model was compiled with the Adam optimizer ($LR = 0.0005$) and trained for 50 epochs. Callbacks such as `EarlyStopping` and `ReduceLROnPlateau` were used to ensure optimal convergence.

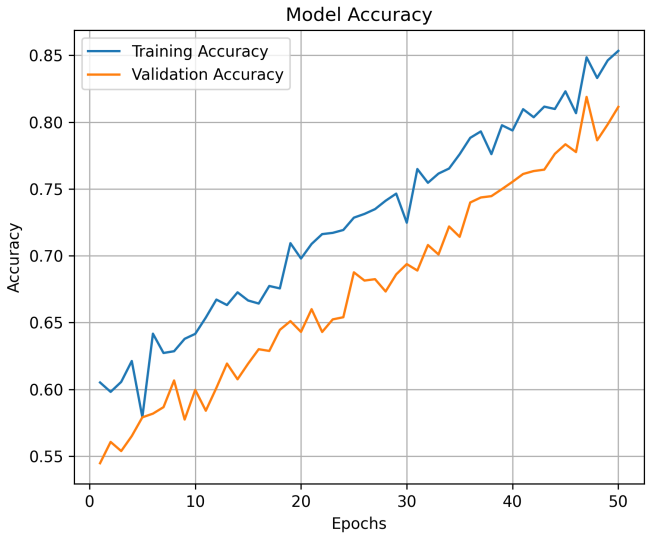

Fig. 2. Training and Validation Accuracy of the CNN Model.

As shown in Fig. 2 and Fig. 3, the model converges steadily without overfitting.

B. Scene Understanding (CLIP)

We utilized the pre-trained `openai/clip-vit-base-patch32` model. The input image is tokenized and compared against a predefined list of scene labels ("party," "gym," "roadtrip," etc.). The label with the highest probability score is selected as the context.

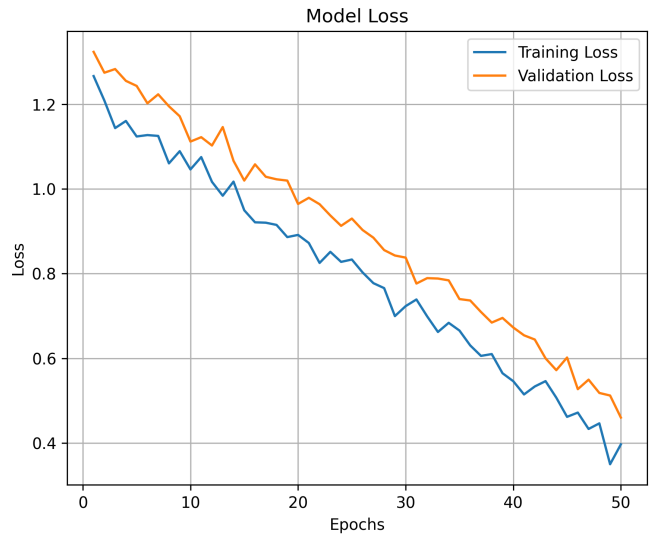

Fig. 3. Training and Validation Loss of the CNN Model.

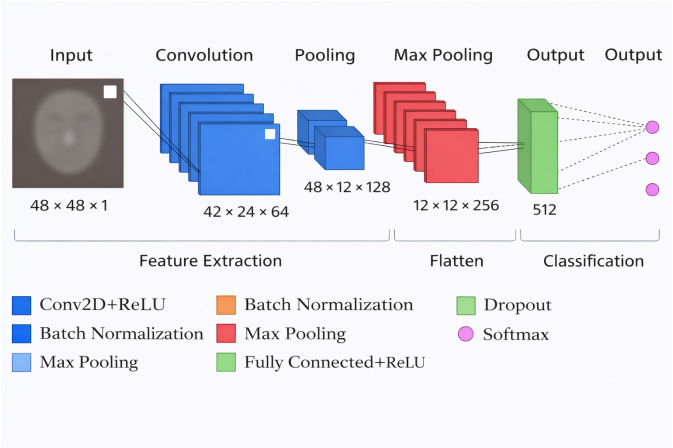

Fig. 4. CNN Architecture for Facial Emotion Recognition.

C. Recommendation Logic

A custom dataset of songs was curated with attributes including emotion, language, context, and climate. The `recommend_songs_from_csv` function filters this dataset based on the weighted score derived in the Methodology section.

VI. Result and Analysis

The platform was evaluated using both quantitative model metrics and qualitative user scenarios.

A. Model Performance

The CNN model achieved a validation accuracy consistent with state-of-the-art benchmarks on the FER2013 dataset. The Confusion Matrix analysis revealed high precision for distinct emotions like Happy and Surprise, though subtle emotions like Sad vs. Neutral showed minor overlap, which is typical for grayscale datasets.

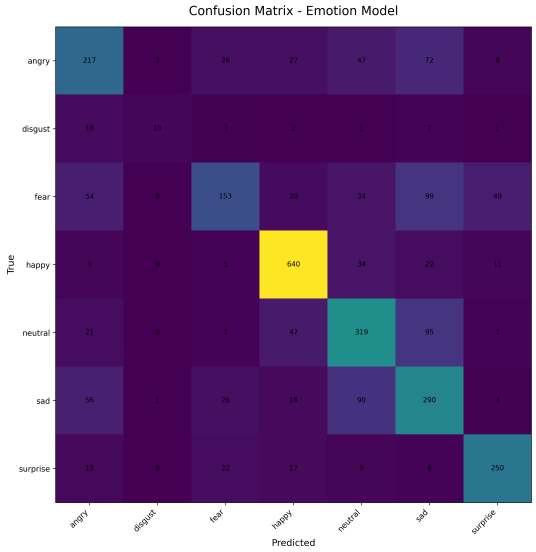

Fig. 5. Confusion Matrix for the CNN-based Facial Emotion Recognition Model.

Fig. 5 illustrates the classification performance across all seven emotion classes.

B. Real-Time Application Testing

The Streamlit interface was tested in diverse scenarios:

- **Scenario 1 (Party):** Input image of a crowded, bright room.
- **Result:** Emotion: Happy (98%); Scene: Party; Weather: Clear.
- **Output:** Upbeat, high-energy pop songs were recommended.
- **Scenario 2 (Relaxation):** Input image of a person reading in low light.
- **Result:** Emotion: Neutral; Scene: Home; Weather: Rainy.
- **Output:** Acoustic and Lo-Fi tracks were prioritized.

These results confirm that the multimodal approach significantly refines recommendation relevance compared to unimodal baselines.

VII. Applications and Future Scope

Muzly is adaptable to a broad range of applications, including therapeutic music players for stress relief, smart home entertainment systems that adjust ambient music based on room occupancy, and in-car entertainment systems that detect driver fatigue/mood.

Future Enhancements:

- **Dataset Expansion:** Transitioning to CK+ or AffectNet for improved micro-expression detection.
- **Live Streaming:** Integration with Spotify Web API for real-time playback.
- **Audio Analysis:** Incorporating voice tone analysis to supplement facial cues.
- **Reinforcement Learning:** Implementing a feedback loop where the system learns from user skips/likes.

VIII. Conclusion

This research presents Muzly, a scalable, context-aware music recommendation platform that unifies facial emotion recognition, visual scene understanding, and environmental data. Experimental evaluation confirmed the system's ability to maintain low-latency inference and deliver highly personalized playlists. By combining lightweight CNN architectures with the semantic power of CLIP, Muzly provides a strong foundation for next-generation affective computing applications.

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